Project Proposal: Fault Localization for JFreeChart using SBFL

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1- Project Goal:

The goal of this project is to find bugs in a Java program called JFreeChart. I will use a technique called **Spectrum-Based Fault Localization (SBFL)**. This technique helps find which part of the code is suspicious and likely has a bug by looking at which parts are used in passing and failing tests.

2- Data and Language

I will use the coverage data provided in the .zip file from D2L. This data has many .txt files showing which Java methods are used by each test. I will use the **Java** programming language to write my tool because it is good for reading files and doing the math needed for this project.

3- Implementation Steps

First, I will write Java code to open and read all the .txt coverage files. Since the data only has true tests, I will manually change some tests to be "failed" tests. Then, my Java code will calculate these four different ways to see which one is best at finding the bug:

- **Tarantula:** This uses a ratio of failed tests vs passed tests. If a method is used a lot in fails but not in passes, the score goes up.

  Suspicious(s) = (fail(s) / totalfail) / ((fail(s) / totalfail) + (pass(s) / totalpass))

- **SBI (Simple Binary Indicator):** A simpler math way to see the percentage of failures.

  Suspicious(s) = fail(s) / (fail(s) + pass(s))

- **Jaccard:** This is a standard formula used in data science to see the "overlap" between the method and the failures.

  Suspicious(s) = fail(s) / (allfailed + pass(s))

- **Ochiai:** This is usually the best one for software bugs. It uses a square root to compare the method coverage to the total failures.

  Suspicious(s) = fail(s) / sqrt(allfailed * (fail(s) + pass(s)))

4- Expected Results:

After the math is done, my program will have a big list of methods and their scores.

- **Sorting Logic:** I will use Collections.sort() in Java with a custom Comparator. This will put the highest suspiciousness score at the top.
- **Final Output:** My program will print a final report. It will show the Method Name and its score for all four formulas. I will compare the results to see if the same method stays at the top of the list for all formulas. This list tells the programmer: "Look at these top 10 methods first to find the bug."